Quantum many-body entanglement and tensor networks

Learning outcomes

- Understand the degree of quantum entanglement in different models
- Rationalize how and when tensor-networks allow solving quantum many-body Hamiltonians

The two limits of many-body Hamiltonains

A Hamiltonian with many-body interactions takes the form

$$H = \sum_{ij} t_{ij} c_i^\dagger c_j + \sum_{ijkl} V_{ijkl} c_i^\dagger c_j c_k^\dagger c_l$$

With a mean field description

$$H \approx \sum_{ij} \bar{t}_{ij} c_i^\dagger c_j + \sum_{ij} \Delta_{ij} c_i c_j$$

Approximate quadratic Hamiltonian
Effective single particle description

Weakly correlated matter

Without a mean field description

????

No good quadratic approximation
Requires exact solutions or numerical

Strongly correlated matter

Fermionic and spin many-body-Hamiltonians

We can have two types of many-body Hamiltonians

Many-body fermionic Hamiltonians

$$H = \sum_{ij} t_{ij} c_i^\dagger c_j + \sum_{ijkl} V_{ijkl} c_i^\dagger c_j c_k^\dagger c_l$$

Many-body spin Hamiltonians

$$H = \sum_{ij} J_{ij} \vec{S}_i \cdot \vec{S}_j$$

Can we systematically map spin and fermionic Hamiltonians?

Fermionic and spin many-body-Hamiltonians

For a single site, the local Hilbert space is similar



Leading to identifying the operators as $c^\dagger \sim S^+$ $c \sim S^-$

However, between different sites, fermionic operators anticommute, whereas spin commute

The right algebra can be recovered adding a properly chosen filling-dependent sign

The equivalence between fermionic and spin Hamiltonians

If we have a generic fermionic Hamiltonian, we can transform it into a spin Hamiltonian with

$$S_i^+ = c_i^\dagger e^{i\pi \sum_{i<j} c_j^\dagger c_j} \qquad S_i^- = c_i e^{-i\pi \sum_{i<j} c_j^\dagger c_j} \qquad S_i^z = c_i^\dagger c_i - \frac{1}{2}$$

Jordan-Wigner transformation

spin = fermion x string

$$S_i^+ \qquad c_i^\dagger \qquad e^{-i\pi \sum_{i<j} c_j^\dagger c_j}$$

This transformation fulfills the algebraic relations of spins and fermions

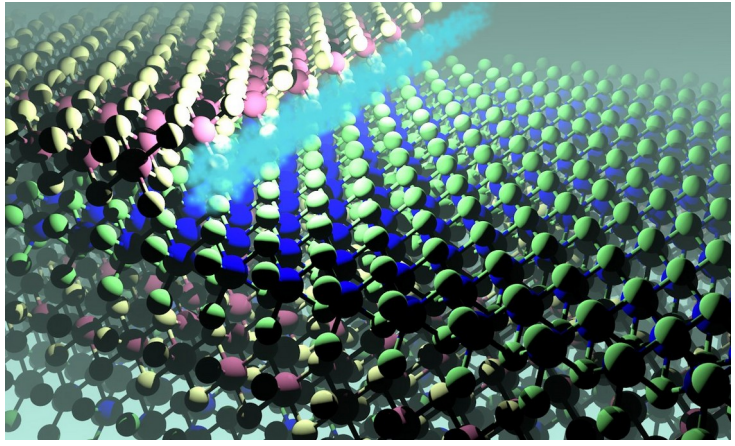
In the following, let us then focus on spin many-body Hamiltonians

The fundamental idea of tensor-networks

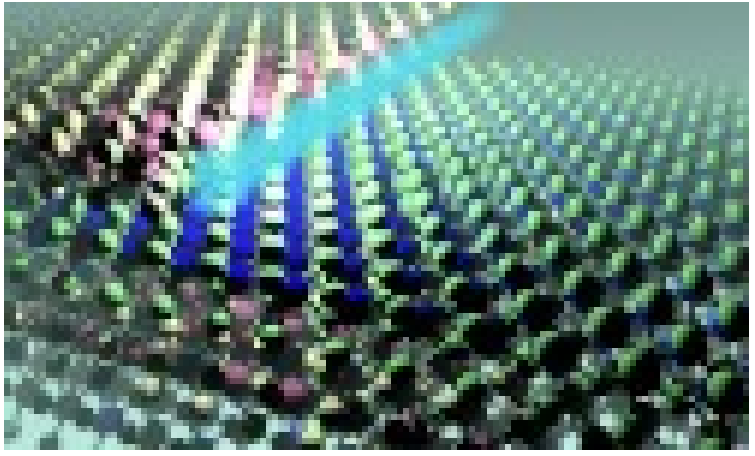
A many-body wavefunction a is a very high dimensional object

$$|\Psi\rangle = \sum c_{s_1,s_2,\ldots,s_L} |s_1, s_2, \ldots s_L\rangle$$

Tensor-networks allow “compressing” all that information in a very efficient way



“True wavefunction”



“Tensor-network wavefunction”

The quantum Heisenberg dimer

What is the ground state of this quantum Hamiltonian?

$$\mathcal{H} = \vec{S}_0 \cdot \vec{S}_1$$

The ground state is unique, and does not break time-reversal

$$|GS\rangle = \frac{1}{\sqrt{2}}[|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle] \qquad \langle \vec{S}_i \rangle = 0$$

The state is maximally entangled

How does this generalize to a macroscopic system?

The quantum many-body problem

Let us go back to a simple many-body problem $\mathcal{H} = \sum_{ij} J_{ij} \vec{S}_i \cdot \vec{S}_j$

And let us imagine that we have L different sites on our system and S=1/2

For example, for L=2 sites the elements of the basis are

$|\uparrow\uparrow\rangle \quad |\uparrow\downarrow\rangle \quad |\downarrow\uparrow\rangle \quad |\downarrow\downarrow\rangle$

For L=3 sites the elements of the basis are

$|\uparrow\uparrow\uparrow\rangle \quad |\uparrow\uparrow\downarrow\rangle \quad |\uparrow\downarrow\uparrow\rangle \quad |\uparrow\downarrow\downarrow\rangle$

$|\downarrow\uparrow\uparrow\rangle \quad |\downarrow\uparrow\downarrow\rangle \quad |\downarrow\downarrow\uparrow\rangle \quad |\downarrow\downarrow\downarrow\rangle$

The quantum many-body problem

Let us go back to a simple many-body problem $\mathcal{H} = \sum_{ij} J_{ij} \vec{S}_i \cdot \vec{S}_j$

And let us imagine that we have L different sites on our system and S=1/2

For L=4 sites, the elements of the basis are

$$|\uparrow\uparrow\uparrow\uparrow\rangle \quad |\uparrow\uparrow\uparrow\downarrow\rangle \quad |\uparrow\uparrow\downarrow\uparrow\rangle \quad |\uparrow\uparrow\downarrow\downarrow\rangle$$
$$|\uparrow\downarrow\uparrow\uparrow\rangle \quad |\uparrow\downarrow\uparrow\downarrow\rangle \quad |\uparrow\downarrow\downarrow\uparrow\rangle \quad |\uparrow\downarrow\downarrow\downarrow\rangle$$
$$|\downarrow\uparrow\uparrow\uparrow\rangle \quad |\downarrow\uparrow\uparrow\downarrow\rangle \quad |\downarrow\uparrow\downarrow\uparrow\rangle \quad |\downarrow\uparrow\downarrow\downarrow\rangle$$
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The quantum many-body problem

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The dimension of the Hilbert space grows as

$$d = 2^L$$

The quantum many-body problem

Let us go back to a simple many-body problem $\mathcal{H} = \sum_{ij} J_{ij} \vec{S}_i \cdot \vec{S}_j$

A typical wavefunction is written as

$$|\Psi\rangle = \sum c_{s_1,s_2,\ldots,s_L} |s_1, s_2, \ldots s_L\rangle$$

We need to determine in total 2^L coefficients

Is there an efficient way of storing so many coefficients?

Matrix product states and tensor networks

The matrix-product state ansatz

For this wavefunction $|\Psi\rangle = \sum c_{s_1,s_2,\ldots,s_L} |s_1, s_2, \ldots s_L\rangle$

Let us imagine to propose a parametrization in this form

$$c_{s_1,s_2,\ldots,s_L} = M_1^{s_1} M_2^{s_2} \ldots . M_L^{s_3}$$

dimension 2^L

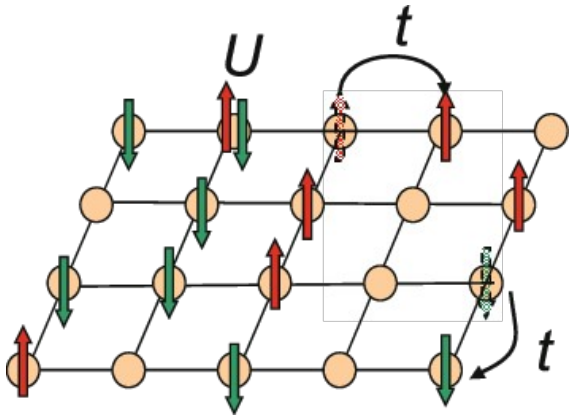
dimension $\sim Lm^2$

(m dimension of the matrix)

Why is this is a good parametrization of a wavefunction?

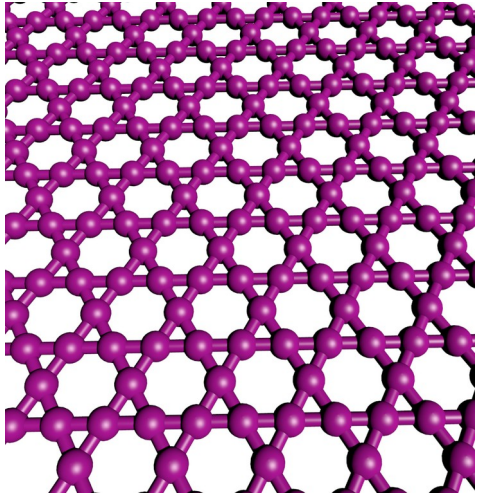
Some paradigmatic quantum materials tackled with tensor networks

Solving the 2D Hubbard model at finite doping



$$H = \sum_{ij,s} t_{ij} c^{\dagger}_{is} c_{js} + \sum_{i} U c^{\dagger}_{i\uparrow} c_{i\uparrow} c^{\dagger}_{i\downarrow} c_{i\downarrow}$$

Solving the 2D Heisenberg model in frustrated lattices

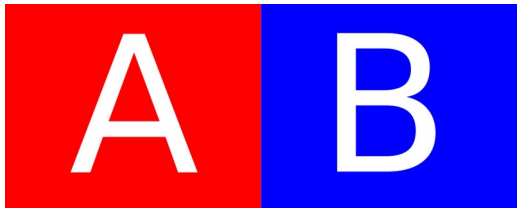


$$\mathcal{H} = \sum_{ij} J_{ij} \vec{S}_i \cdot \vec{S}_j$$

Entanglement in a many-body wavefunction

If a state can be written as

$$|\Psi\rangle = |\Psi_A\rangle \otimes |\Psi_B\rangle$$



Then we say that it is not entangled

For example the following states are not entangled

$$|\Psi\rangle = |\uparrow\downarrow\rangle \qquad |\Psi\rangle = (|\uparrow\rangle_1 + |\downarrow\rangle_1) \otimes (|\uparrow\rangle_2 + |\downarrow\rangle_2)$$

But this state is entangled $$|\Psi\rangle = \frac{1}{\sqrt{2}}[|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle]$$

How can we know, and quantify, how entangled an state is?

The entanglement entropy

We can quantify the entanglement in a system through the entanglement entropy

Define the density matrix $$\rho = |\Psi\rangle \langle\Psi|$$

Trace over one subsystem (reduced density matrix) $$\rho_A = \mathrm{Tr}_B(\rho)$$

Defining the entropy of the state $$S_A = -\mathrm{Tr}(\rho_A \log \rho_A)$$

For a product state $$|\Psi\rangle = |\Psi_A\rangle \otimes |\Psi_B\rangle$$

the entanglement entropy is identically zero

The entanglement entropy of a gaped 1d model

Take a dimerized Heisenberg model of the form $H = \sum_n J \vec{S}_{2n} \cdot \vec{S}_{2n+1}$



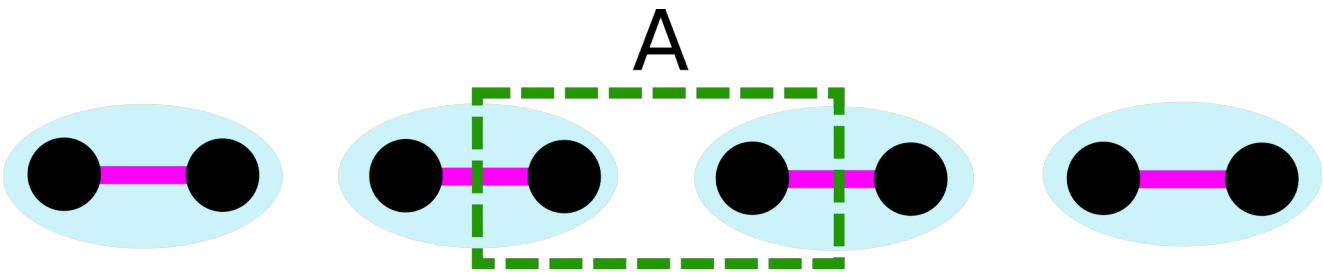
The many-body wavefunction is just a product of singlet between dimers

$$|\Psi\rangle \sim \prod_{\otimes n} (|\uparrow_{2n}\downarrow_{2n+1}\rangle - |\downarrow_{2n}\uparrow_{2n+1}\rangle)$$

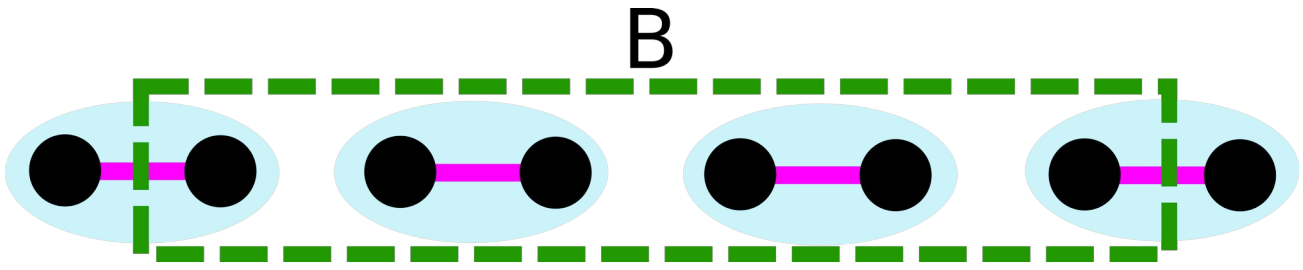
What is the entanglement entropy when making a bipartition?

The entanglement entropy of a gaped 1d model

Let us first take a small partition A, and compute the entanglement entropy



The entanglement entropy stems from the number of singlets that are cut
If the bipartition is bigger, the number of singlets that are cut remains the same

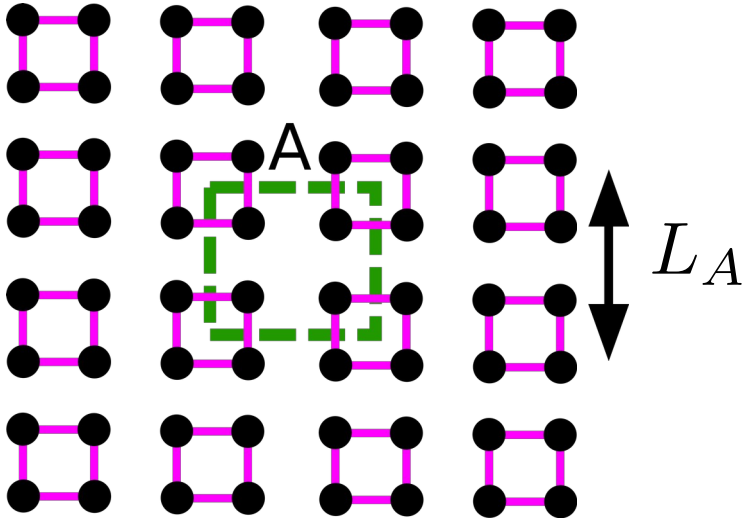


The entanglement entropy does not depend on the size of the partition $S_A \sim S_B \sim 2\log(2)$

The entanglement entropy of a gaped 2d model

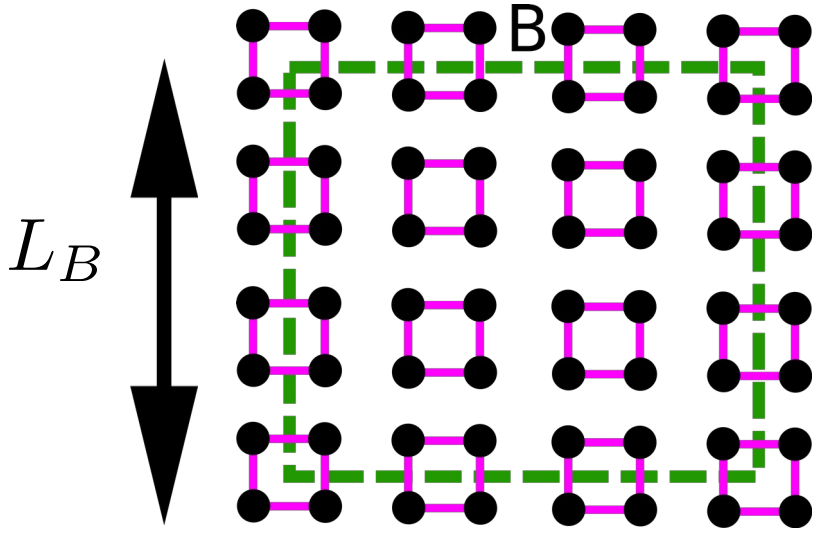
Let us consider the analogous Heisenberg model dimerized in 2D $H = \sum_{ij} J_{ij} \vec{S}_i \cdot \vec{S}_j$

Partition A



$S_B > S_A$

Partition B

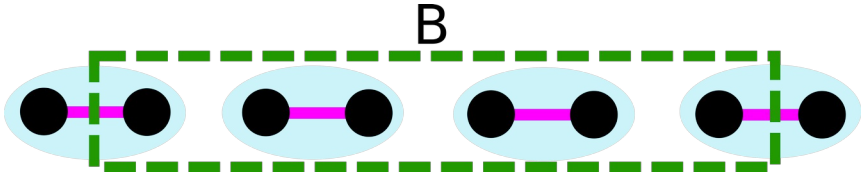


$S_A \sim L_A \sim \partial A$

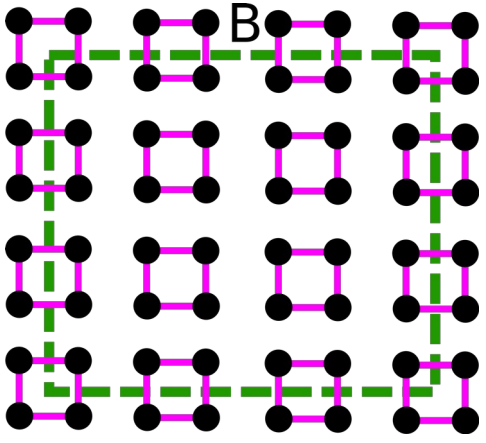
$S_B \sim L_B \sim \partial B$

Entanglement entropy and the area law

The ground state of gapped systems follows the area law, meaning that the entanglement entropy of a subsystem depends of the length of the boundary



In 1D, the entropy is independent of the system size

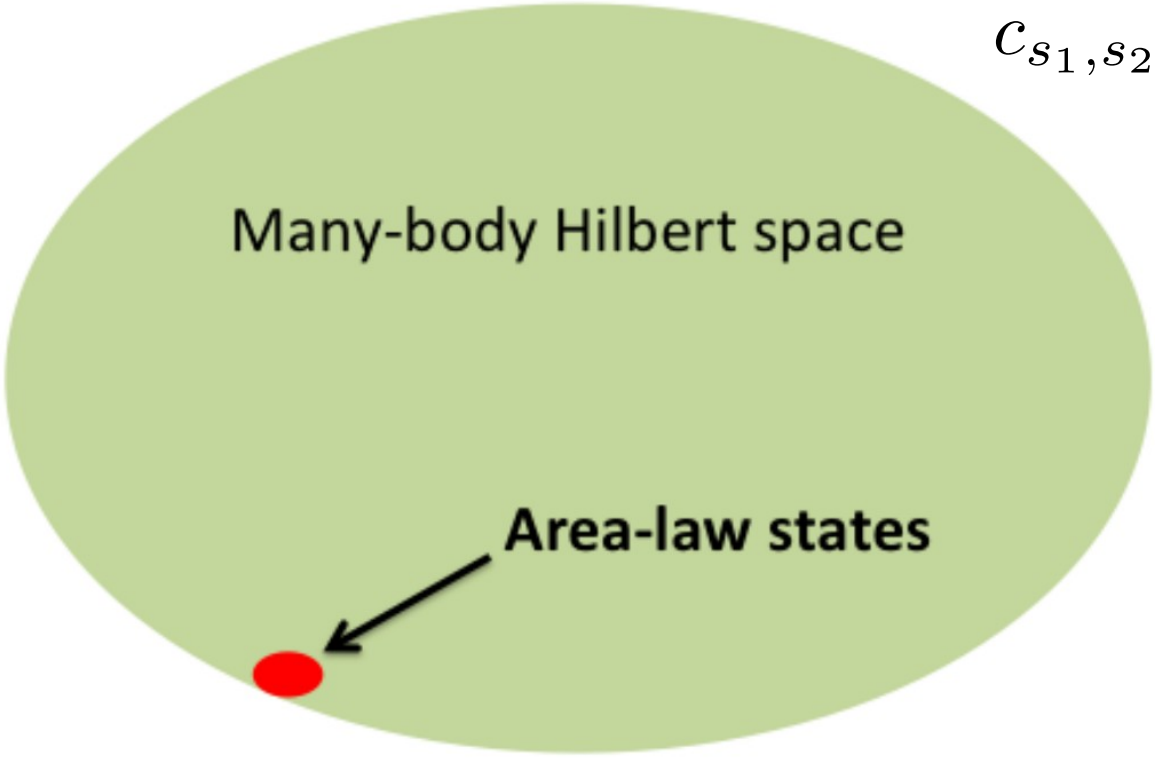


In 2D, it grows linearly with the size of the subsystem

Matrix product states have an entanglement entropy controlled by the size of the matrices

$$c_{s_1,s_2,....,s_L} = M_1^{s_1} M_2^{s_2} ....M_L^{s_3}$$

Matrix product states and area law states



$$c_{s_1,s_2,....,s_L} = M_1^{s_1} M_2^{s_2} ....M_L^{s_3}$$

$$M \sim m \times m \text{ matrix}$$

Entanglement entropy of an MPS

$$S_A \sim \log(m)$$

The matrix-product state ansatz

- This ansatz enforces a maximum amount of entanglement entropy in the state $S \sim \log m$
- One-dimensional problems, have ground states that can be captured with this ansatz

$$c_{s_1,s_2,....,s_L} = M_1^{s_1} M_2^{s_2} ....M_L^{s_3}$$

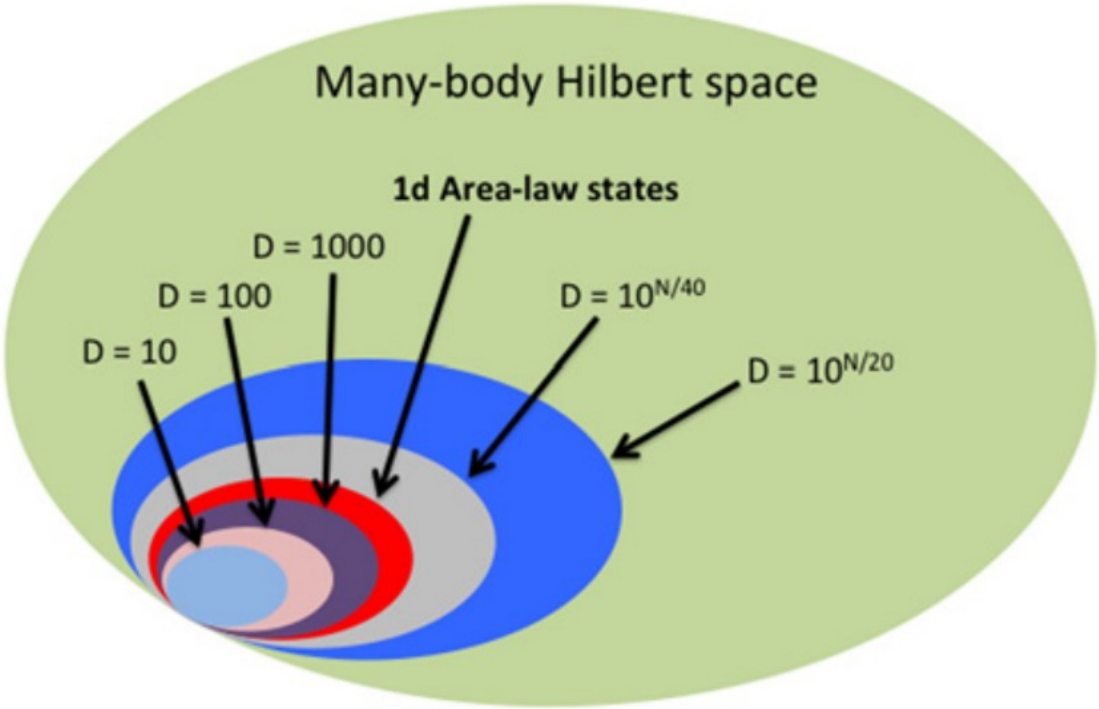
This ansatz can be generalized for time-evolution, excited states, open systems.

When do matrix product states fail?

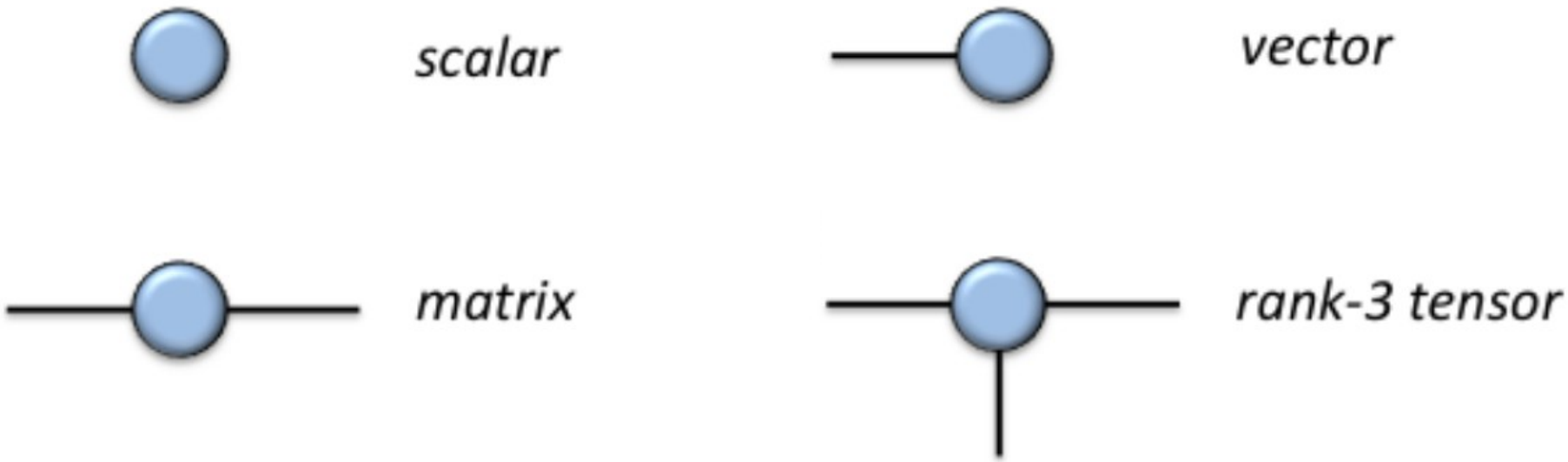
- This ansatz enforces a maximum amount of entanglement entropy in the state $S \sim \log m$
- If the states have too much entanglement, MPS do not capture the state properly
  - Time-evolution to long times
  - Many-body problems above 1D
  - Highly excited states
  - Far from equilibrium states

When do MPS fail?

Sketch of the space parametrized with bond dimension D

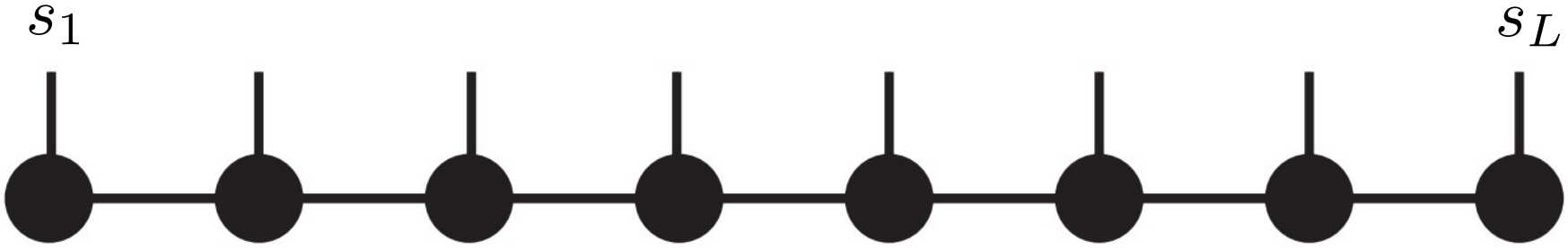


The graphical representation of matrix product states



Tensor-network operations can be easily described with diagrams

The matrix product state representation

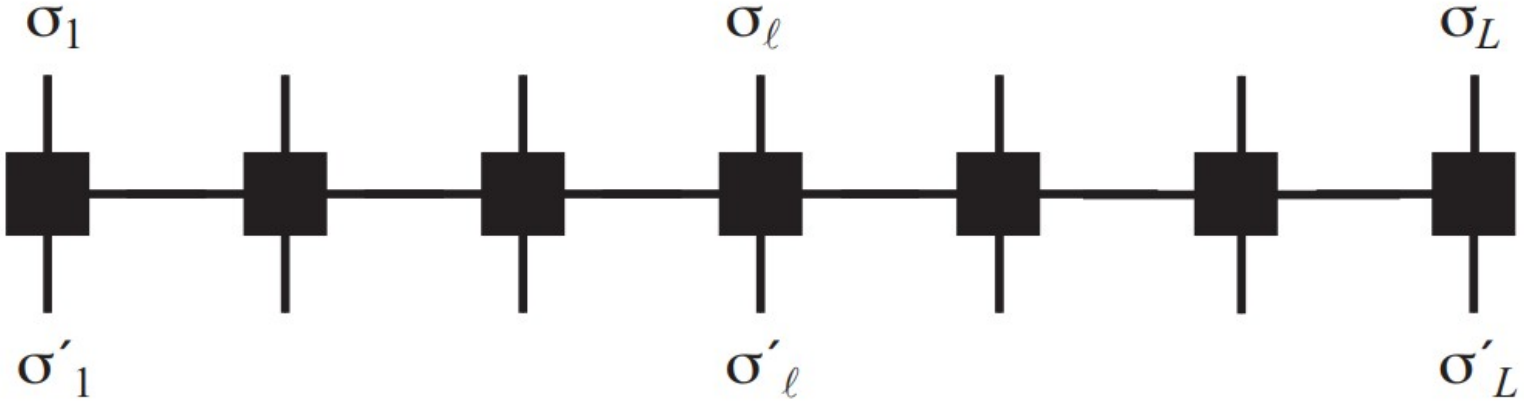


$$|\Psi\rangle = \sum c_{s_1,s_2,\ldots,s_L} |s_1, s_2, \ldots s_L\rangle$$

$$c_{s_1,s_2,\ldots,s_L} = M_1^{s_1} M_2^{s_2} \ldots M_L^{s_3}$$

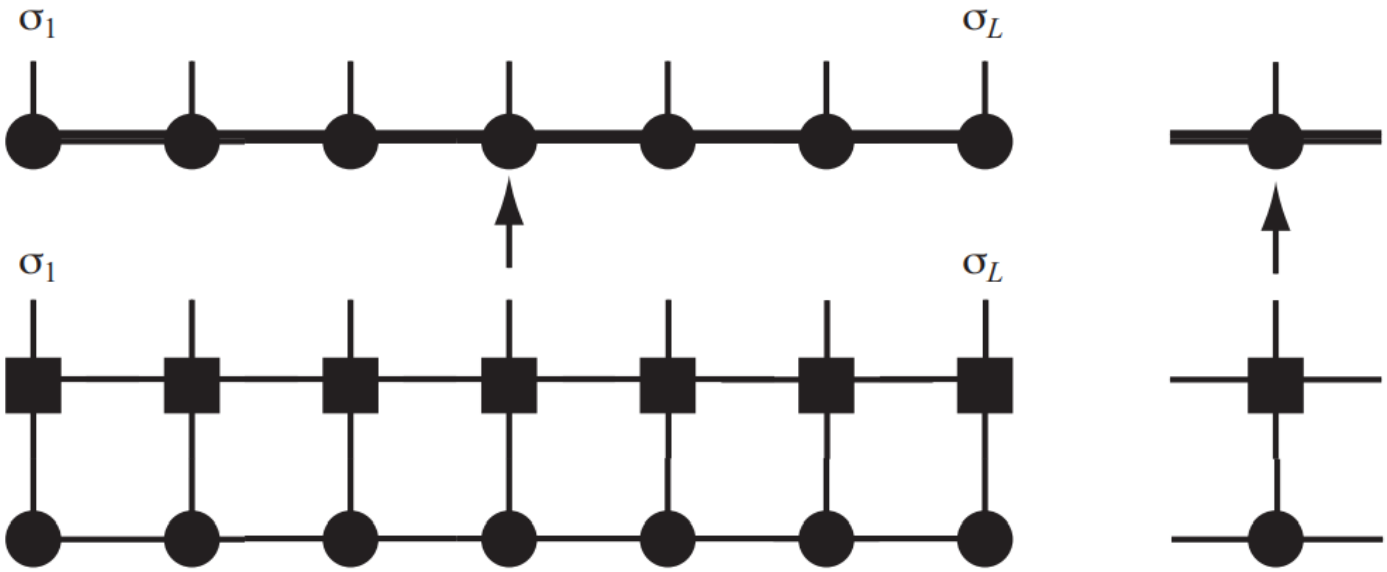
Matrix product operators

Operators can be represented in an analogous form



Operator state product

Products of operators and states can be represented graphically



Ground state calculations

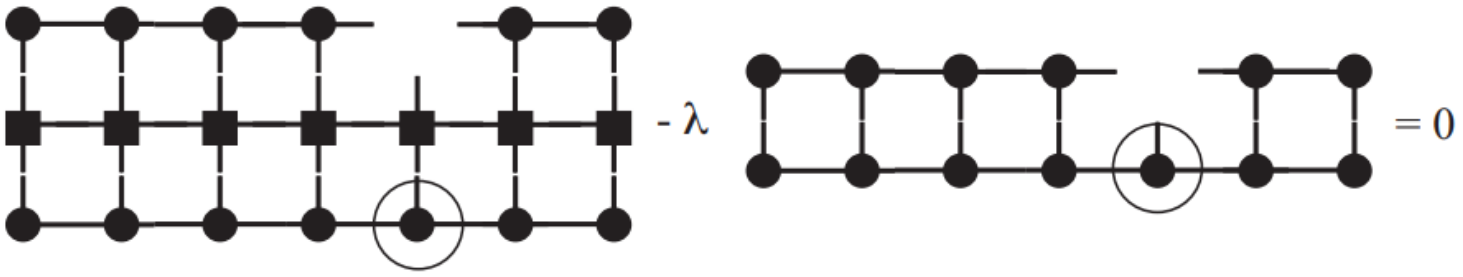
To compute a ground state, we just have to minimize

$$E = \frac{\langle \Psi | H | \Psi \rangle}{\langle \Psi | \Psi \rangle}$$

$$E = \langle \Psi | H | \Psi \rangle - \lambda \langle \Psi | \Psi \rangle$$

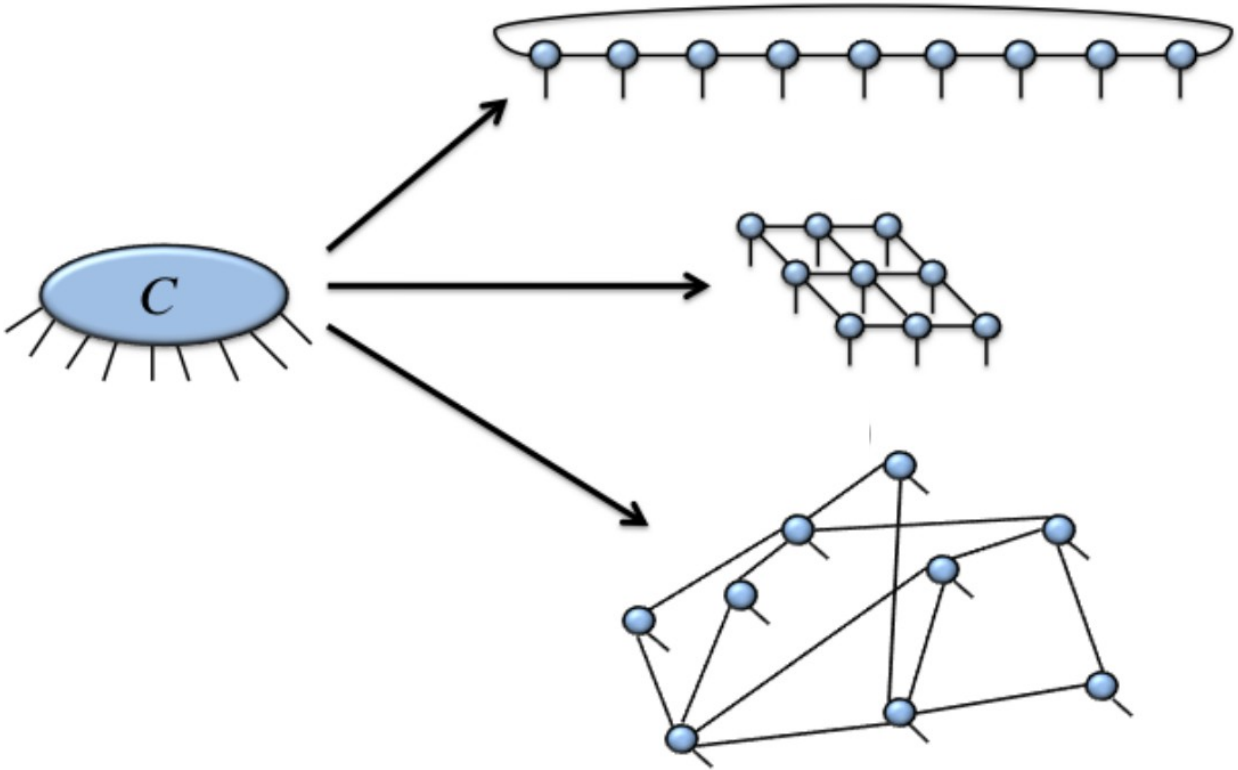
This can be done by minimizing the energy with respect to each matrix

$$\frac{\delta E}{\delta M} = 0$$



This algorithm is known as the density-matrix renormalization group

Beyond matrix-product states

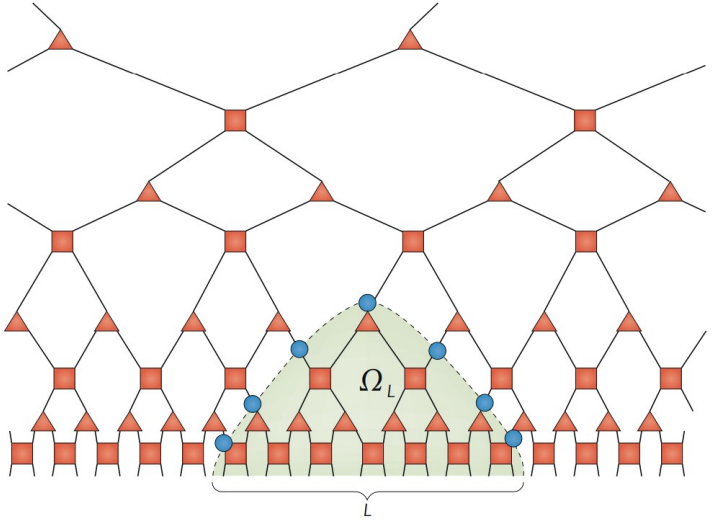


The same tensor can be represented with widely different tensor networks

Beyond matrix-product states

Tensor networks can be extended to deal with higher dimensional/critical systems

Multiscale renormalization ansatz



Projected-entangled pair states

